

R2.1 How much? The amount of chemical change

Limiting and excess reactants, theoretical and percentage yield

Learning outcomes

By the end of this section you should be able to:

- (R2.1.3) Identify the limiting and excess reactants from given data.
- (R2.1.3) Distinguish between the theoretical yield and the experimental yield and suggest errors which may cause the experimental yield to be either higher or lower than the theoretical yield.
- (R2.1.4) Solve problems involving reacting quantities, limiting and excess reactants, theoretical, experimental and percentage yields.

Recall the recipe for pancakes we have been discussing and representing as a chemical equation (**Figure 1**).

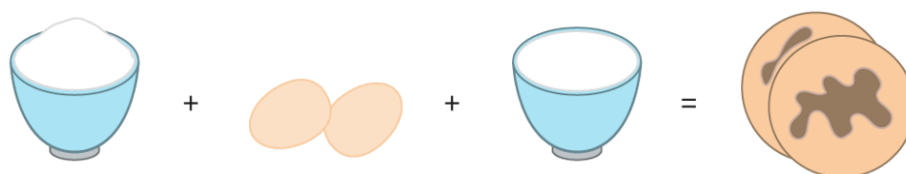
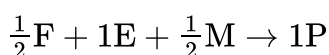


Figure 1. Pancake recipe equation.

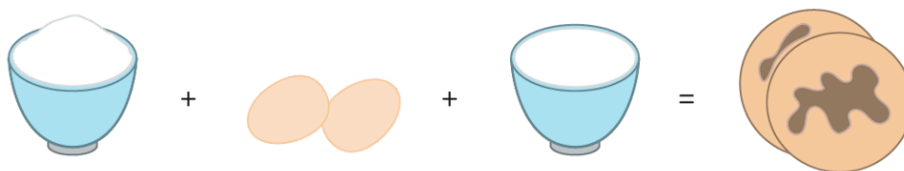
Previously we discussed halving the recipe to make one pancake because there was only one egg:



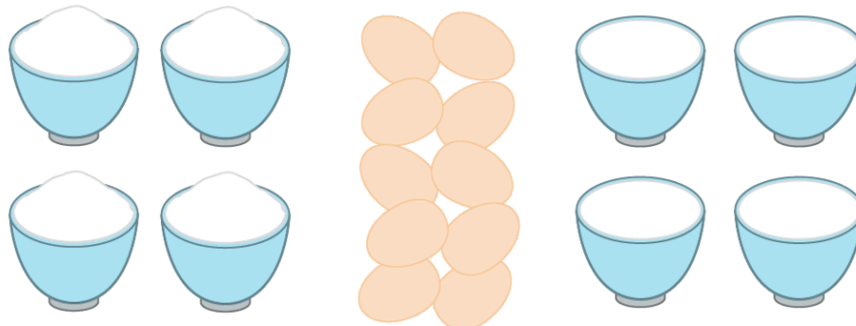
What would happen if there was one egg, but one cup of flour and one cup of milk? Could you make two pancakes?

No, it does not matter if there is excess of other ingredients – the egg will still limit how many pancakes you can make. As the limiting ingredient, the egg decides the number of pancakes you can make, i.e. the yield of pancakes (**Figure 2**).

1 cup of flour + 2 eggs + 1 cup of milk = 2 pancakes



Provided with 4 cups of flour + 10 eggs + 4 cups of milk



We can make 8 pancakes + 2 eggs left over

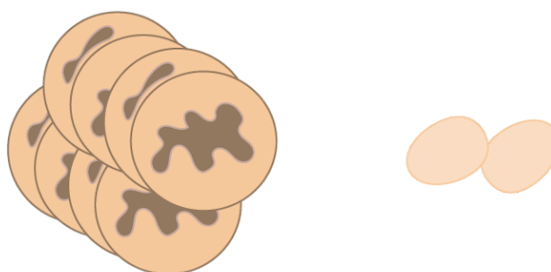


Figure 2. Limiting and excess ingredients.

 More information for figure 2

The image illustrates a recipe scenario to explain the concept of limiting ingredients. At the top, there is a line of text: "1 cup of flour + 2 eggs + 1 cup of milk = 2 pancakes." Below this, three images are arranged horizontally: a bowl of flour, two eggs, and a bowl of milk, with two pancakes at the end, symbolizing the result.

In the middle section, the text reads: "Provided with 4 cups of flour + 10 eggs + 4 cups of milk." This is depicted with four bowls of flour, ten eggs, and four bowls of milk arranged in three rows aligned with their respective text.

At the bottom, the text states: "We can make 8 pancakes + 2 eggs left over." There is an illustration showing eight pancakes stacked together and two leftover eggs, emphasizing that the eggs are the limiting factor in the scenario.

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How can we apply this same idea to chemical reactions?

Limiting and excess reactants

Many chemical reactions will also have imperfect amounts of reactants which do not fit the required ratio.

You may have more of one reactant than the other and it simply does not all get used up; it will still remain at the end of the reaction – this reactant is said to be in excess. The limiting reactant is the reactant which is always fully used up. In the example above involving the pancake recipe we identified that the limiting reactant, the egg, was what decided the number of pancakes being produced. This is a key feature of limiting reactants; they always decide the amount of product, known as the yield.

The excess reactant is the one which is remaining after the reaction has taken place. Can you identify the excess reactant in **Figure 3**?

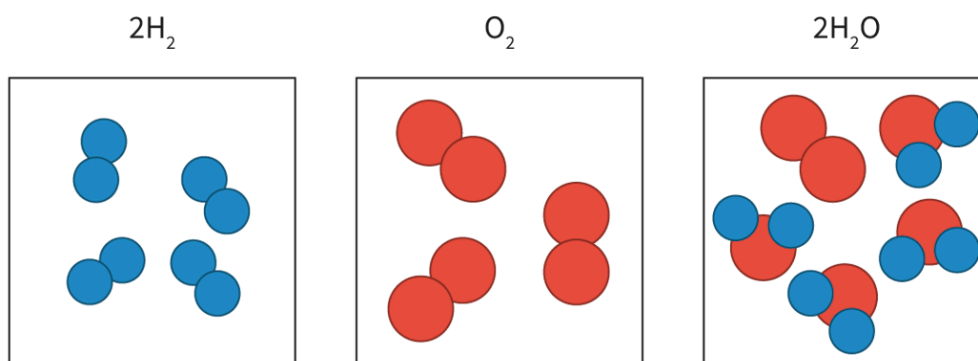


Figure 3. Limiting and excess reactants.

 More information for figure 3

The diagram consists of three sections, each illustrating molecular sets and reactions. The first section displays three pairs of blue circles, each representing molecules of H_2 (hydrogen). The middle section shows two pairs and a single red circle, indicating O_2 (oxygen) molecules. The third section on the right displays molecules forming, showing red and blue circles connected, symbolizing H_2O (water) post-reaction. The image illustrates the transition from reactants to product, with leftover molecules illustrating excess reactant details.

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To determine the limiting reactant, you should:

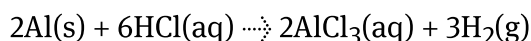
1. Calculate the number of moles of each reactant.
2. Divide the number of moles of each reactant by their coefficient in the reaction. Compare the two numbers – the smaller number is the limiting reactant.
3. Use the original moles of limiting reactant, and the mole ratio to calculate the yield of the product in focus.

The species in the reaction can be present in any phase and you should be prepared to convert solids, liquids and gases into moles using the data you are given.

Worked example 1

10.0 g of aluminium, Al, is added to 50.0 cm³ of a 0.100 mol dm⁻³

solution of hydrochloric acid. Aluminium chloride and hydrogen gas are formed according to this equation:



Determine the limiting reactant.

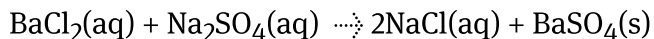
Solution steps	Calculations
Step 1: Determine the limiting reactant by first converting both reactants to moles.	Moles of Al Al mass = 10.0 g Al $M = 26.98 \text{ g mol}^{-1}$ Al moles = $\frac{m}{M}$ = 0.371 mol Moles of HCl HCl volume = 50.0 cm³ = 0.0500 dm³

Solution steps	Calculations
	HCl concentration: 0.100 mol dm⁻³ HCl moles = $C \times V$ = 0.00500 mol
Step 2: Divide the mole values for both by their coefficients.	Mole ratio: 2Al : 6HCl 2 : 6 Moles of Al Al moles: 0.371 mol Coefficient: 2 $\frac{0.371}{2} = 0.186 \text{ mol}$ Moles of HCl HCl moles: 0.00500 mol Coefficient: 6 $\frac{0.005}{6} = 0.000833 \text{ mol}$ The smaller number of moles is this is the limiting reactant.

Once the limiting reactant has been determined it's possible to calculate the yield using the moles and the original mole ratio.

Worked example 2

100.0 cm³ of a 0.5000 mol dm⁻³ solution of BaCl₂ is added to 50.00 cm³ of a 0.1000 mol dm⁻³ solution of Na₂SO₄. A precipitate of BaSO₄ is formed according to this balanced chemical equation:



1. Determine the limiting reactant.
2. Determine the yield of BaSO_4 produced in grams.

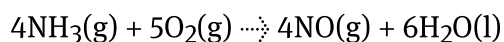
Solution steps	Calculations
Step 1: Determine the limiting reactant by first converting both reactants to moles.	Moles of BaCl_2 $\text{BaCl}_2 \text{ volume} = 100.00 \text{ cm}^3$ $= 0.1000 \text{ dm}^3$ BaCl_2 concentration = 0.5000 $\text{BaCl}_2 \text{ moles} = C \times V$ $= 0.05000 \text{ mol}$ Moles of Na_2SO_4 $\text{Na}_2\text{SO}_4 \text{ volume: } 50.00 \text{ cm}^3 =$ $\text{Na}_2\text{SO}_4 \text{ concentration} = 0.10$ $\text{Na}_2\text{SO}_4 \text{ moles: } C \times V = 0.0050$
Step 2: Divide the mole values for both by their coefficients.	Mole ratio $\text{BaCl}_2 : \text{Na}_2\text{SO}_4$ $1 : 1$ Both values are divided by their The smaller number of moles is Na_2SO_4 ($5.000 \times 10^{-3} \text{ mol}$) so
Step 3: Calculate experimental yield using the mole ratio of the limiting reactant and the product in focus.	To calculate theoretical yield of B Mole ratio $\text{Na}_2\text{SO}_4 : \text{BaSO}_4$ $1 : 1$

Solution steps	Calculations
	$5.000 \times 10^{-3} : x$ $x = 5.000 \times 10^{-3} \text{ mol BaSO}_4$ $\text{BaSO}_4 \text{ moles: } 5.000 \times 10^{-3} \text{ mol}$ $\text{BaSO}_4 \text{ } M = 233.40 \text{ g mol}^{-1}$ Mass of BaSO_4 $n = \frac{m}{M}$ $5.000 \times 10^{-3} = \frac{m}{233.40}$ $m = 1.167 \text{ g}$

If you know which reactant is limiting and which one is in excess, then it's also possible to determine how much excess reactant would remain after the reaction.

Worked example 3

A reaction occurs between ammonia and oxygen which forms nitrogen monoxide and water:



100.0 grams of ammonia were reacted with 120.0 grams of oxygen.

(a) Identify which reagent is the limiting reagent.

(b) Calculate the mass of excess reactant which remains when the reaction is completed.

Part (a)

Solution steps	Calculations
Step 1: Determine the limiting reactant by first converting both reactants to moles.	Moles of NH_3 $\text{NH}_3 \text{ mass} = 100.0 \text{ gNH}_3$ $\text{NH}_3 \text{ moles} = \frac{m}{M}$ $= 5.870 \text{ mol}$ Moles of O_2 $\text{O}_2 \text{ mass} = 120.0 \text{ gO}_2$ $\text{O}_2 \text{ moles} = \frac{m}{M}$ $= 3.750 \text{ mol}$
Step 2: Divide the mole values for both by their coefficients.	Mole ratio $4\text{NH}_3 : 5\text{O}_2$ Moles of NH_3 $\text{NH}_3 \text{ moles: } 5.870 \text{ mol}$ Coefficient: 4 $\frac{5.870}{4} = 1.470 \text{ mol}$ Moles of O_2 $\text{O}_2 \text{ moles: } 3.750 \text{ mol}$ Coefficient: 5 $\frac{3.750}{5} = 0.7500 \text{ mol}$ The smaller number of moles is for O_2, so this is the limiting reactant.

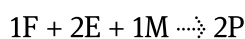
Part (b)

Solution steps	Calculations
Step 1: Calculate the mass of excess reactant which remains after the reaction.	First determine the moles of excess up using mole ratios: Mole ratio $4\text{NH}_3 : 5\text{O}_2$ $x : 3.750$ $x = 3.000 \text{ mol NH}_3 \text{ will be used in the reaction}$ Moles of NH_3 present in 100.00 g of NH_3 $\text{Excess moles of NH}_3 = 5.870 - 3.000$ $= 2.870 \text{ mol}$ So 2.870 mol NH_3 remain after the reaction Mass of excess NH_3 $\text{NH}_3 \text{ moles} = \frac{m}{M}$ $= 2.870 \text{ mol}$ $\text{NH}_3 \text{ } M = 17.04 \text{ g mol}^{-1}$ Hence $\text{NH}_3 \text{ mass} = 48.90 \text{ g}$

Understanding and controlling which reactant is the limiting reactant allows for industry to produce very specific amounts of products. Why do you think the limiting reactant is usually the more expensive one when this is applied in industry?

Theoretical and percentage yield

Consider the original pancake recipe again:



The recipe should yield two pancakes but when pouring the pancake batter, half of it was accidentally spilled. Although there should have been two pancakes yielded in theory, there was actually only one. Half the yield was lost, i.e. 50%.

The theoretical yield that we calculate represents the amount that should be produced if a reaction is fully carried out under ideal conditions. How is theoretical yield calculated?

Based on our understanding that all of the limiting reactant is used up, we can use that to determine what is the maximum product that could be formed from that amount of reactant. The limiting reactant is always what decides the theoretical yield (**Figure 4**).

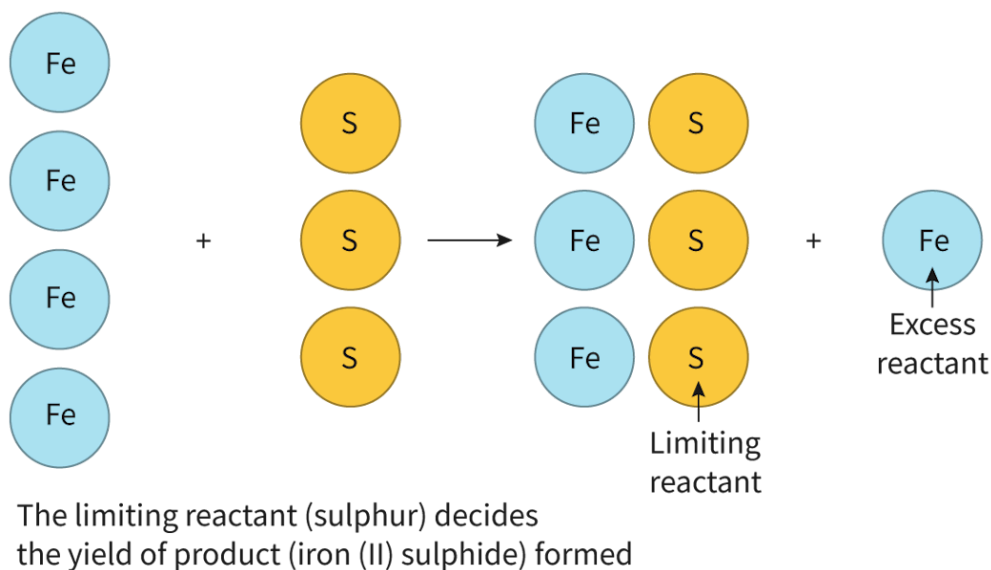


Figure 4. The limiting reactant (sulfur) decides the yield of product (iron(II) sulfide) formed.

 More information for figure 4

The diagram visually represents a chemical reaction between iron (Fe) and sulfur (S) to form iron(II) sulfide. On the left side, there are four blue circles labeled 'Fe' representing iron atoms, and three yellow circles labeled 'S' for sulfur atoms. An arrow points to the reaction products on the right, showing three pairs of Fe and S combined, indicating the formation of iron sulfide. One iron atom is left unreacted, labeled as 'Excess reactant'. A yellow circle marked 'Limiting reactant' near sulfur indicates sulfur is the limiting reactant that decides the yield of the product. This illustrates that the reaction stops when the sulfur is used up.

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We can use the limiting reactant to determine the theoretical yield which should be produced but in reality, there are many factors that can cause there to be less product than expected, and result in a lower experimental yield.

Some of these reasons are:

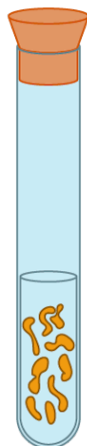
- **Loss of product:** some of the product may evaporate or simply be spilled. It may also be lost when isolating or purifying the product.
- **Incomplete reaction:** there may be impurities present or the reaction may not go to completion due to insufficient time.
- **Alternative reactions:** an unexpected side reaction may occur which results in a different product.
- **One or more of the reactants is impure.**

These things could also occur in the pancake making process. A lower yield obtained could be caused by using flour that was not pure (impurity may undergo a different reaction), not mixing the batter properly (an incomplete reaction), or not scraping clean the batter from the mixing bowl (loss of product).

Making connections

Another reason a reaction may not produce the expected theoretical yield is if the reaction is reversible. If the position of equilibrium lies to the left for that reaction then the reactants will not stay converted into products. You will learn more about this in [subtopic R2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/).

It is also possible for the experimental yield to be greater than the theoretical yield. This can occur when there are impurities present in the product, such as the presence of moisture or a solvent if the drying of a precipitate or solid product is incomplete. It may also occur if the product undergoes subsequent reactions with its surroundings, such as oxidising or decomposing over time (**Figure 5**).



Incomplete drying of a precipitate formed may lead to a higher percentage yield than expected

Figure 5. Incomplete drying can affect the percentage yield.

 More information for figure 5

The image depicts a test tube filled partially with light blue liquid and containing an orange-colored precipitate. The test tube is topped with an orange stopper. Below the test tube, textual information explains that incomplete drying of a precipitate formed may lead to a higher percentage yield than expected. The illustration is intended to demonstrate the concept of incomplete drying affecting chemical yield.

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A more convenient way of comparing the theoretical and experimental yield is as a percentage yield. The percentage yield represents the efficiency of the conversion of reactants into products. This is calculated using the formula:

$$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

Study skills

The formula for percentage yield is not provided for you in the DP chemistry data booklet so it is important that you can recall it.

Worked example 4

A biochemist was performing a reaction in which she expected to produce 17.89 kg of a product, but actually only produced 11.75 kg of product.

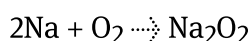
Calculate the percentage yield and suggest reasons why the experimental yield did not achieve 100%.

Solution steps	Calculations
Step 1: Identify theoretical and experimental yield from question.	Theoretical yield = 17.89 kg Experimental yield = 11.75 kg
Step 2: Substitute the values into the equation.	$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$ $= \frac{11.75}{17.89} \times 100\%$ $= 65.68\%$ Possible reasons why experimental yield is less than theoretical yield: • The reaction is reversible so not all reactants are converted to products. • An alternative reaction may have occurred. • Some of the product may have been lost during isolation.

Questions will often require you to first calculate the theoretical yield. You should do this using the value which the limiting reactant would produce.

Worked example 5

Determine the percentage yield for the reaction between 3.74 g of Na and excess O₂ if 5.34 g of Na₂O₂ is recovered:

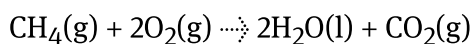


Solution steps	Calculations
Step 1: Determine the limiting reactant and calculate the maximum mass of product which could theoretically be obtained. The question states that oxygen is in excess so sodium must be the limiting reactant.	Moles of Na Na mass = 3.74g Na $M = 22.99 \text{ g mol}^{-1}$ Na moles = $\frac{m}{M}$ = 0.163 mol Mole ratio 2Na : Na₂O₂ 0.163 : $xx = 0.0800 \text{ mol Na}_2\text{O}_2$ Mass of Na₂O₂ Na₂O₂ moles = 0.0815 mol Na₂O₂ $M = 77.98 \text{ g mol}^{-1}$ Mass of Na₂O₂ = $n \times M$ = 6.36 g
Step 2: Identify theoretical and experimental yield.	Theoretical yield = 6.36 g Experimental yield = 5.34 g
Step 3: Substitute the values into the equation.	percentage yield = $\frac{\text{experimental}}{\text{theoretical}}$ = $\frac{5.34}{6.36} \times 100$ = 84.0%

Remember that reactants and products can be solids, liquids, gas or aqueous so be prepared to convert each of them to moles.

Worked example 6

Calculate the percentage yield of water from the combustion of methane if 19.8 g of water was obtained experimentally from a mixture of 20.0 dm³ of methane and 42.0 dm³ of oxygen gas. Assume that the reaction was carried out at STP.



Solution steps	Calculations
Step 1: Determine the limiting reactant and calculate the maximum mass of product which could theoretically be obtained. The question does not specify which reactant is the limiting reactant so that must first be determined.	CH_4 volume = 20.0 dm³ $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ $\text{CH}_4 \text{ moles} = \frac{V}{V_m}$ $= 0.880 \text{ mol}$ O_2 volume = 42.0 dm³ $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ $\text{O}_2 \text{ moles} = \frac{V}{V_m}$ $= 1.85 \text{ mol}$ Mole ratio is 1 : 2 $\text{CH}_4(\text{g}) : 2\text{O}_2(\text{g})$ If O_2 was the limiting reactant: $x \text{ mol CH}_4 : 1$ $x = 0.930 \text{ mol}$ of CH_4 would be required, but we only have 0.880 mol available. If CH_4 was the limiting reactant: 0.880 mol CH_4 requires $2 \times 0.880 = 1.76 \text{ mol}$ of O_2. We have 1.85 mol available. CH_4 must be the limiting reactant and therefore the theoretical yield of the product.

Solution steps	Calculations
Step 2: Identify theoretical and experimental yield.	Experimental yield = 19.8g To calculate theoretical yield of H₂O Mole ratio CH₄ : 2H₂O 1 : 2 0.880 : x $x = 1.76$ mol H₂O moles = 1.76 mol H₂O $M = 18.02$ g mol⁻¹ Mass of H₂O = $n \times M$ = 31.7g Remember that the density of water is 1 g cm⁻³ so the mass and volume for water is quite easy. This (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) booklet.
Step 3: Substitute the values into the equation.	$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$ $= \frac{19.8}{31.7} \times 100\%$ $= 62.4\%$

Scientists in industry work to maximise the percentage yield of a reaction to ensure that profits are high and there is very little wastage of raw materials. An example of this is in the pharmaceutical industry where a small difference in percentage purity can translate into millions of dollars.

The next activity will allow you to further explore the concept of limiting and excess reactants and to practise calculating them, the yield, and the percentage yield.